Residential exposure to pesticides and childhood leukaemia: a systematic review and meta-analysis.

OBJECTIVE: To conduct a systematic review of published studies on the association between residential/household/domestic exposure to pesticides and childhood leukaemia, and to provide a quantitative estimate of the risk. METHODS: Publications in English were searched in MEDLINE (1966-31 December 2009) and from the reference list of identified publications. Extraction of relative risk (RR) estimates was performed independently by 2 authors using predefined inclusion criteria. Meta-rate ratio estimates (mRR) were calculated according to fixed and random-effect models. Separate analyses were conducted after stratification for exposure time windows, residential exposure location, biocide category and type of leukaemia. RESULTS: RR estimates were extracted from 13 case-control studies published between 1987 and 2009. Statistically significant associations with childhood leukaemia were observed when combining all studies (mRR: 1.74, 95% CI: 1.37-2.21). Exposure during and after pregnancy was positively associated with childhood leukaemia, with the strongest risk for exposure during pregnancy (mRR: 2.19, 95% CI: 1.92-2.50). Other stratifications showed the greatest risk estimates for indoor exposure (mRR: 1.74, 95% CI: 1.45-2.09), for exposure to insecticides (mRR: 1.73, 95% CI: 1.33-2.26) as well as for acute non-lymphocytic leukaemia (ANLL) (mRR: 2.30, 95% CI: 1.53-3.45). Outdoor exposure and exposure of children to herbicides (after pregnancy) were not significantly associated with childhood leukaemia (mRR: 1.21, 95% CI: 0.97-1.52; mRR: 1.16, 95% CI: 0.76-1.76, respectively). CONCLUSIONS: Our findings support the assumption that residential pesticide exposure may be a contributing risk factor for childhood leukaemia but available data were too scarce for causality ascertainment. It may be opportune to consider preventive actions, including educational measures, to decrease the use of pesticides for residential purposes and particularly the use of indoor insecticides during pregnancy.